United States Patent Application Publication	20250287271
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	SRIVASTAVA; Akash et al.

VOICE CALL CONTINUITY IMPROVEMENTS FROM WLAN TO WWAN

Abstract

Certain aspects of the present disclosure provide techniques for improving voice call continuity from a wireless local area network (WLAN) to a wireless wide area network (WWAN). An example method performed by a user equipment (UE) includes performing, while a session is active on a first radio access technology (RAT) network, reselection to a second RAT network from a third RAT network, performing a handover to the second RAT network, after detecting a condition impacting the session on the first RAT network, and resuming the session on the second RAT network.

Inventors: **SRIVASTAVA; Akash (Hyderabad, IN), ALLU; Siva Krishna (Hyderabad, IN), PALANISAMY; Santhana (Hyderabad, IN), KONAPPA; Srinivasa Ragimakalahally (Bangalore, IN), PRATURI; Upendra Ram (Hyderabad, IN), THALAPATHI; Jayaraj (Coimbatore, IN), BHATIA; Varun (Ghaziabad, IN)**

Applicant: **QUALCOMM Incorporated** (San Diego, CA)

Family ID: **96950058**

Appl. No.: **18/596337**

Filed: **March 05, 2024**

Publication Classification

Int. Cl.: **H04W36/00** (20090101); **H04W36/14** (20090101); **H04W36/30** (20090101); **H04W36/36** (20090101); **H04W88/06** (20090101)

U.S. Cl.:

CPC **H04W36/00226** (20230501); **H04W36/1446** (20230501); **H04W36/304** (20230501); **H04W36/362** (20230501); H04W88/06 (20130101)

Background/Summary

BACKGROUND

Field of the Disclosure

[0001] Aspects of the present disclosure relate to wireless communications, and more particularly, to techniques for improving voice call continuity from a wireless local area network (WLAN) to a wireless wide area network (WWAN).

Description of Related Art

[0002] Wireless communications systems are widely deployed to provide various telecommunication services such as telephony, video, data, messaging, broadcasts, or other similar types of services. These wireless communications systems may employ multiple-access technologies capable of supporting communications with multiple users by sharing available wireless communications system resources with those users.

[0003] Although wireless communications systems have made great technological advancements over many years, challenges still exist. For example, complex and dynamic environments can still attenuate or block signals between wireless transmitters and wireless receivers. Accordingly, there is a continuous desire to improve the technical performance of wireless communications systems, including, for example: improving speed and data carrying capacity of communications, improving efficiency of the use of shared communications mediums, reducing power used by transmitters and receivers while performing communications, improving reliability of wireless communications, avoiding redundant transmissions and/or receptions and related processing, improving the coverage area of wireless communications, increasing the number and types of devices that can access wireless communications systems, increasing the ability for different types of devices to intercommunicate, increasing the number and type of wireless communications mediums available for use, and the like. Consequently, there exists a need for further improvements in wireless communications systems to overcome the aforementioned technical challenges and others.

SUMMARY

[0004] One aspect provides a method for wireless communications at a user equipment (UE). The method includes performing, while a session is active on a first radio access technology (RAT) network, reselection to a second RAT network from a third RAT network; performing a handover to the second RAT network, after detecting a condition impacting the session on the first RAT network; and resuming the session on the second RAT network.

[0005] Other aspects provide: an apparatus operable, configured, or otherwise adapted to perform any one or more of the aforementioned methods and/or those described elsewhere herein; a non-transitory, computer-readable media comprising instructions that, when executed by one or more processors of an apparatus, cause the apparatus to perform the aforementioned methods as well as those described elsewhere herein; a computer program product embodied on a computer-readable storage medium comprising code for performing the aforementioned methods as well as those described elsewhere herein; and/or an apparatus comprising means for performing the aforementioned methods as well as those described elsewhere herein. By way of example, an apparatus may comprise a processing system, a device with a processing system, or processing systems cooperating over one or more networks.

[0006] The following description and the appended figures set forth certain features for purposes of illustration.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0007] The appended figures depict certain features of the various aspects described herein and are not to be considered limiting of the scope of this disclosure.

[0008] FIG. **1** depicts an example wireless communications network.

[0009] FIG. **2** depicts an example disaggregated base station architecture.

[0010] FIG. **3** depicts aspects of an example base station and an example user equipment.

[0011] FIGS. **4**A, **4**B, **4**C, and **4**D depict various example aspects of data structures for a wireless communications network.

[0012] FIG. **5** is a call flow diagram including operations that illustrate a call muting issue.

[0013] FIG. **6** is a call flow diagram including operations that illustrate a call dropping issue.

[0014] FIG. **7** depicts a process flow for communications in a network for alleviating the call muting issue.

[0015] FIG. **8** depicts a process flow for communications in a network or alleviating the call dropping issue.

[0016] FIG. **9** depicts a method for wireless communications.

[0017] FIG. **10** depicts aspects of an example communications device.

DETAILED DESCRIPTION

[0018] Aspects of the present disclosure provide apparatuses, methods, processing systems, and computer-readable mediums for improving voice call continuity from a wireless local area network (WLAN) to a wireless wide area network (WWAN).

[0019] For example, in some cases, a user equipment (UE) may be capable of performing a voice call over a WLAN radio access technology (RAT) network, such as WiFi RAT network. When doing so, however, there may be some scenarios in which a quality or strength of a connection to the WLAN deteriorates or drops below a threshold, which may significantly impact a quality of the voice call. When the quality or strength of the connection drops below the threshold, the UE may be configured to perform a handover from the WLAN RAT network to a WWAN RAT network, such as a long term evolution (LTE) RAT network, to reestablish and continue the voice call. Generally, this process may be known as evolved packet system fall back (EPSFB). In other words, when the voice call on the WLAN RAT network is interrupted (e.g., due to the quality or signal strength of the connection deteriorating below the threshold), EPSFB involves the UE falling back to the LTE RAT network to continue the voice call.

[0020] However, in some cases, the LTE RAT network may have a lower priority relative to another WWAN RAT network, such as a fifth generation new radio (5G NR) RAT network. As a result of the 5G NR RAT network having a relatively higher priority, the UE may be required to first perform a handover from the WLAN RAT network (e.g., WiFi RAT network) to the 5G NR RAT network before being redirected by the 5G NR RAT network to the LTE RAT network. In other words, the UE may first have to perform a handover to the 5G NR RAT network, receive a redirection command from the 5G RAT network, and then perform another handover to the LTE RAT network.

[0021] Having to perform a handover to the 5G NR RAT network, receive the redirection command, and then perform a second handover to the LTE RAT network consumes a significant amount of time. Moreover, during the period of time when these handovers are occurring, the voice call will remain muted (e.g., no voice data is able to be communicated, which leads to poor user experience at the UE.

[0022] Additionally, in some cases, there may be scenarios in which the handover to the 5G NR RAT network is rejected and the UE is not redirected to the LTE RAT network. When the handover to the 5G NR RAT network is rejected, the UE may be forced to stay on the WLAN RAT network until the connection deteriorates to a certain level that the voice call drops, again leading to poor user experience at the UE.

[0023] Accordingly aspects of the present disclosure provide techniques to help alleviate the call muting and call dropping issues that may arise when performing a voice call over a WLAN RAT network. For example, in some cases, to help alleviate the call muting issue, these techniques may involve the UE performing a reselection procedure during the voice call to reselect the LTE RAT network prior to a WLAN connection deteriorating below a threshold and irrespective of a relative priority assigned to the LTE RAT network. This may allow the UE to skip performing the handover to the 5G NR RAT network when the connection to the WLAN network deteriorates below the threshold and instead directly perform the handover to the LTE RAT network, thereby reducing the amount of time it takes to handover to the LTE RAT network and to reestablish the voice call.
[0024] Additionally, to help alleviate the call dropping issues, the techniques provided herein may involve the UE de-prioritizing a 5G RAT network and attempting a handover to an LTE network entity when the UE receives a handover reject message from a 5G network entity on the 5G RAT network. This may avoid the scenario in which the UE is effectively stuck on the WLAN RAT network and unable to handover to the LTE network entity due to the 5G RAT network having priority over the LTE RAT network. For example, by de-prioritizing a 5G RAT network, the UE may then be free to attempt a handover to an LTE network entity to reestablish and continue the voice call.

Introduction to Wireless Communications Networks

[0025] The techniques and methods described herein may be used for various wireless communications networks. While aspects may be described herein using terminology commonly associated with 3G, 4G, and/or 5G wireless technologies, aspects of the present disclosure may likewise be applicable to other communications systems and standards not explicitly mentioned herein.
[0026] FIG. **1** depicts an example of a wireless communications network **100**, in which aspects described herein may be implemented.
[0027] Generally, wireless communications network **100** includes various network entities (alternatively, network elements or network nodes). A network entity is generally a communications device and/or a communications function performed by a communications device (e.g., a user equipment (UE), a base station (BS), a component of a BS, a server, etc.). For example, various functions of a network as well as various devices associated with and interacting with a network may be considered network entities. Further, wireless communications network **100** includes terrestrial aspects, such as ground-based network entities (e.g., BSs **102**), and non-terrestrial aspects, such as satellite **140** and aircraft **145**, which may include network entities on-board (e.g., one or more BSs) capable of communicating with other network elements (e.g., terrestrial BSs) and user equipments.
[0028] In the depicted example, wireless communications network **100** includes BSs **102**, UEs **104**, and one or more core networks, such as an Evolved Packet Core (EPC) **160** and 5G Core (5GC) network **190**, which interoperate to provide communications services over various communications links, including wired and wireless links.
[0029] FIG. **1** depicts various example UEs **104**, which may more generally include: a cellular phone, smart phone, session initiation protocol (SIP) phone, laptop, personal digital assistant (PDA), satellite radio, global positioning system, multimedia device, video device, digital audio player, camera, game console, tablet, smart device, wearable device, vehicle, electric meter, gas pump, large or small kitchen appliance, healthcare device, implant, sensor/actuator, display, internet of things (IoT) devices, always on (AON) devices, edge processing devices, or other similar devices. UEs **104** may also be referred to more generally as a mobile device, a wireless device, a wireless communications device, a station, a mobile station, a subscriber station, a mobile subscriber station, a mobile unit, a subscriber unit, a wireless unit, a remote unit, a remote device, an access terminal, a mobile terminal, a wireless terminal, a remote terminal, a handset, and others.
[0030] BSs **102** wirelessly communicate with (e.g., transmit signals to or receive signals from) UEs

104 via communications links **120**. The communications links **120** between BSs **102** and UEs **104** may include uplink (UL) (also referred to as reverse link) transmissions from a UE **104** to a BS **102** and/or downlink (DL) (also referred to as forward link) transmissions from a BS **102** to a UE **104**. The communications links **120** may use multiple-input and multiple-output (MIMO) antenna technology, including spatial multiplexing, beamforming, and/or transmit diversity in various aspects.

[0031] BSs **102** may generally include: a NodeB, enhanced NodeB (eNB), next generation enhanced NodeB (ng-eNB), next generation NodeB (gNB or gNodeB), access point, base transceiver station, radio base station, radio transceiver, transceiver function, transmission reception point, and/or others. Each of BSs **102** may provide communications coverage for a respective geographic coverage area **110**, which may sometimes be referred to as a cell, and which may overlap in some cases (e.g., small cell **102**′ may have a coverage area **110**′ that overlaps the coverage area **110** of a macro cell). A BS may, for example, provide communications coverage for a macro cell (covering relatively large geographic area), a pico cell (covering relatively smaller geographic area, such as a sports stadium), a femto cell (relatively smaller geographic area (e.g., a home)), and/or other types of cells.

[0032] While BSs **102** are depicted in various aspects as unitary communications devices, BSs **102** may be implemented in various configurations. For example, one or more components of a base station may be disaggregated, including a central unit (CU), one or more distributed units (DUs), one or more radio units (RUs), a Near-Real Time (Near-RT) RAN Intelligent Controller (RIC), or a Non-Real Time (Non-RT) RIC, to name a few examples. In another example, various aspects of a base station may be virtualized. More generally, a base station (e.g., BS **102**) may include components that are located at a single physical location or components located at various physical locations. In examples in which a base station includes components that are located at various physical locations, the various components may each perform functions such that, collectively, the various components achieve functionality that is similar to a base station that is located at a single physical location. In some aspects, a base station including components that are located at various physical locations may be referred to as a disaggregated radio access network architecture, such as an Open RAN (O-RAN) or Virtualized RAN (VRAN) architecture. FIG. **2** depicts and describes an example disaggregated base station architecture.

[0033] Different BSs **102** within wireless communications network **100** may also be configured to support different radio access technologies, such as 3G, 4G, and/or 5G. For example, BSs **102** configured for 4G LTE (collectively referred to as Evolved Universal Mobile Telecommunications System (UMTS) Terrestrial Radio Access Network (E-UTRAN)) may interface with the EPC **160** through first backhaul links **132** (e.g., an S**1** interface). BSs **102** configured for 5G (e.g., 5G NR or Next Generation RAN (NG-RAN)) may interface with 5GC **190** through second backhaul links **184**. BSs **102** may communicate directly or indirectly (e.g., through the EPC **160** or 5GC **190**) with each other over third backhaul links **134** (e.g., X**2** interface), which may be wired or wireless.

[0034] Wireless communications network **100** may subdivide the electromagnetic spectrum into various classes, bands, channels, or other features. In some aspects, the subdivision is provided based on wavelength and frequency, where frequency may also be referred to as a carrier, a subcarrier, a frequency channel, a tone, or a subband. For example, 3GPP currently defines Frequency Range **1** (FR**1**) as including 410 MHz-7125 MHz, which is often referred to (interchangeably) as “Sub-6 GHz”. Similarly, 3GPP currently defines Frequency Range **2** (FR**2**) as including 24,250 MHz-71,000 MHz, which is sometimes referred to (interchangeably) as a “millimeter wave” (“mmW” or “mmWave”). In some cases, FR**2** may be further defined in terms of sub-ranges, such as a first sub-range FR**2-1** including 24,250 MHz-52,600 MHz and a second sub-range FR**2-2** including 52,600 MHz-71,000 MHz. A base station configured to communicate using mm Wave/near mm Wave radio frequency bands (e.g., a mmWave base station such as BS **180**) may utilize beamforming (e.g., **182**) with a UE (e.g., **104**) to improve path loss and range.

[0035] The communications links **120** between BSs **102** and, for example, UEs **104**, may be through one or more carriers, which may have different bandwidths (e.g., 5, 10, 15, 20, 100, 400, and/or other MHz), and which may be aggregated in various aspects. Carriers may or may not be adjacent to each other. Allocation of carriers may be asymmetric with respect to DL and UL (e.g., more or fewer carriers may be allocated for DL than for UL).

[0036] Communications using higher frequency bands may have higher path loss and a shorter range compared to lower frequency communications. Accordingly, certain base stations (e.g., **180** in FIG. **1**) may utilize beamforming **182** with a UE **104** to improve path loss and range. For example, BS **180** and the UE **104** may each include a plurality of antennas, such as antenna elements, antenna panels, and/or antenna arrays to facilitate the beamforming. In some cases, BS **180** may transmit a beamformed signal to UE **104** in one or more transmit directions **182′**. UE **104** may receive the beamformed signal from the BS **180** in one or more receive directions **182″**. UE **104** may also transmit a beamformed signal to the BS **180** in one or more transmit directions **182″**. BS **180** may also receive the beamformed signal from UE **104** in one or more receive directions **182′**. BS **180** and UE **104** may then perform beam training to determine the best receive and transmit directions for each of BS **180** and UE **104**. Notably, the transmit and receive directions for BS **180** may or may not be the same. Similarly, the transmit and receive directions for UE **104** may or may not be the same.

[0037] Wireless communications network **100** further includes a Wi-Fi AP **150** in communication with Wi-Fi stations (STAs) **152** via communications links **154** in, for example, a 2.4 GHz and/or 5 GHz unlicensed frequency spectrum.

[0038] Certain UEs **104** may communicate with each other using device-to-device (D2D) communications link **158**. D2D communications link **158** may use one or more sidelink channels, such as a physical sidelink broadcast channel (PSBCH), a physical sidelink discovery channel (PSDCH), a physical sidelink shared channel (PSSCH), a physical sidelink control channel (PSCCH), and/or a physical sidelink feedback channel (PSFCH).

[0039] EPC **160** may include various functional components, including: a Mobility Management Entity (MME) **162**, other MMEs **164**, a Serving Gateway **166**, a Multimedia Broadcast Multicast Service (MBMS) Gateway **168**, a Broadcast Multicast Service Center (BM-SC) **170**, and/or a Packet Data Network (PDN) Gateway **172**, such as in the depicted example. MME **162** may be in communication with a Home Subscriber Server (HSS) **174**. MME **162** is the control node that processes the signaling between the UEs **104** and the EPC **160**. Generally, MME **162** provides bearer and connection management.

[0040] Generally, user Internet protocol (IP) packets are transferred through Serving Gateway **166**, which itself is connected to PDN Gateway **172**. PDN Gateway **172** provides UE IP address allocation as well as other functions. PDN Gateway **172** and the BM-SC **170** are connected to IP Services **176**, which may include, for example, the Internet, an intranet, an IP Multimedia Subsystem (IMS), a Packet Switched (PS) streaming service, and/or other IP services.

[0041] BM-SC **170** may provide functions for MBMS user service provisioning and delivery. BM-SC **170** may serve as an entry point for content provider MBMS transmission, may be used to authorize and initiate MBMS Bearer Services within a public land mobile network (PLMN), and/or may be used to schedule MBMS transmissions. MBMS Gateway **168** may be used to distribute MBMS traffic to the BSs **102** belonging to a Multicast Broadcast Single Frequency Network (MBSFN) area broadcasting a particular service, and/or may be responsible for session management (start/stop) and for collecting eMBMS related charging information.

[0042] 5GC **190** may include various functional components, including: an Access and Mobility Management Function (AMF) **192**, other AMFs **193**, a Session Management Function (SMF) **194**, and a User Plane Function (UPF) **195**. AMF **192** may be in communication with Unified Data Management (UDM) **196**.

[0043] AMF **192** is a control node that processes signaling between UEs **104** and 5GC **190**. AMF

192 provides, for example, quality of service (QoS) flow and session management.

[0044] Internet protocol (IP) packets are transferred through UPF **195**, which is connected to the IP Services **197**, and which provides UE IP address allocation as well as other functions for 5GC **190**. IP Services **197** may include, for example, the Internet, an intranet, an IMS, a PS streaming service, and/or other IP services.

[0045] In various aspects, a network entity or network node can be implemented as an aggregated base station, as a disaggregated base station, a component of a base station, an integrated access and backhaul (IAB) node, a relay node, a sidelink node, to name a few examples.

[0046] FIG. **2** depicts an example disaggregated base station **200** architecture. The disaggregated base station **200** architecture may include one or more central units (CUs) **210** that can communicate directly with a core network **220** via a backhaul link, or indirectly with the core network **220** through one or more disaggregated base station units (such as a Near-Real Time (Near-RT) RAN Intelligent Controller (RIC) **225** via an E**2** link, or a Non-Real Time (Non-RT) RIC **215** associated with a Service Management and Orchestration (SMO) Framework **205**, or both). A CU **210** may communicate with one or more distributed units (DUs) **230** via respective midhaul links, such as an F**1** interface. The DUs **230** may communicate with one or more radio units (RUs) **240** via respective fronthaul links. The RUs **240** may communicate with respective UEs **104** via one or more radio frequency (RF) access links. In some implementations, the UE **104** may be simultaneously served by multiple RUs **240**.

[0047] Each of the units, e.g., the CUs **210**, the DUs **230**, the RUs **240**, as well as the Near-RT RICs **225**, the Non-RT RICs **215** and the SMO Framework **205**, may include one or more interfaces or be coupled to one or more interfaces configured to receive or transmit signals, data, or information (collectively, signals) via a wired or wireless transmission medium. Each of the units, or an associated processor or controller providing instructions to the communications interfaces of the units, can be configured to communicate with one or more of the other units via the transmission medium. For example, the units can include a wired interface configured to receive or transmit signals over a wired transmission medium to one or more of the other units. Additionally or alternatively, the units can include a wireless interface, which may include a receiver, a transmitter or transceiver (such as a radio frequency (RF) transceiver), configured to receive or transmit signals, or both, over a wireless transmission medium to one or more of the other units.

[0048] In some aspects, the CU **210** may host one or more higher layer control functions. Such control functions can include radio resource control (RRC), packet data convergence protocol (PDCP), service data adaptation protocol (SDAP), or the like. Each control function can be implemented with an interface configured to communicate signals with other control functions hosted by the CU **210**. The CU **210** may be configured to handle user plane functionality (e.g., Central Unit-User Plane (CU-UP)), control plane functionality (e.g., Central Unit-Control Plane (CU-CP)), or a combination thereof. In some implementations, the CU **210** can be logically split into one or more CU-UP units and one or more CU-CP units. The CU-UP unit can communicate bidirectionally with the CU-CP unit via an interface, such as the E**1** interface when implemented in an O-RAN configuration. The CU **210** can be implemented to communicate with the DU **230**, as necessary, for network control and signaling.

[0049] The DU **230** may correspond to a logical unit that includes one or more base station functions to control the operation of one or more RUs **240**. In some aspects, the DU **230** may host one or more of a radio link control (RLC) layer, a medium access control (MAC) layer, and one or more high physical (PHY) layers (such as modules for forward error correction (FEC) encoding and decoding, scrambling, modulation and demodulation, or the like) depending, at least in part, on a functional split, such as those defined by the 3rd Generation Partnership Project (3GPP). In some aspects, the DU **230** may further host one or more low PHY layers. Each layer (or module) can be implemented with an interface configured to communicate signals with other layers (and modules) hosted by the DU **230**, or with the control functions hosted by the CU **210**.

[0050] Lower-layer functionality can be implemented by one or more RUs **240**. In some deployments, an RU **240**, controlled by a DU **230**, may correspond to a logical node that hosts RF processing functions, or low-PHY layer functions (such as performing fast Fourier transform (FFT), inverse FFT (iFFT), digital beamforming, physical random access channel (PRACH) extraction and filtering, or the like), or both, based at least in part on the functional split, such as a lower layer functional split. In such an architecture, the RU(s) **240** can be implemented to handle over the air (OTA) communications with one or more UEs **104**. In some implementations, real-time and non-real-time aspects of control and user plane communications with the RU(s) **240** can be controlled by the corresponding DU **230**. In some scenarios, this configuration can enable the DU(s) **230** and the CU **210** to be implemented in a cloud-based RAN architecture, such as a vRAN architecture.
[0051] The SMO Framework **205** may be configured to support RAN deployment and provisioning of non-virtualized and virtualized network elements. For non-virtualized network elements, the SMO Framework **205** may be configured to support the deployment of dedicated physical resources for RAN coverage requirements which may be managed via an operations and maintenance interface (such as an O**1** interface). For virtualized network elements, the SMO Framework **205** may be configured to interact with a cloud computing platform (such as an open cloud (O-Cloud) **290**) to perform network element life cycle management (such as to instantiate virtualized network elements) via a cloud computing platform interface (such as an O**2** interface). Such virtualized network elements can include, but are not limited to, CUs **210**, DUs **230**, RUs **240** and Near-RT RICs **225**. In some implementations, the SMO Framework **205** can communicate with a hardware aspect of a 4G RAN, such as an open eNB (O-eNB) **211**, via an O**1** interface. Additionally, in some implementations, the SMO Framework **205** can communicate directly with one or more RUs **240** via an O**1** interface. The SMO Framework **205** also may include a Non-RT RIC **215** configured to support functionality of the SMO Framework **205**.
[0052] The Non-RT RIC **215** may be configured to include a logical function that enables non-real-time control and optimization of RAN elements and resources, Artificial Intelligence/Machine Learning (AI/ML) workflows including model training and updates, or policy-based guidance of applications/features in the Near-RT RIC **225**. The Non-RT RIC **215** may be coupled to or communicate with (such as via an Al interface) the Near-RT RIC **225**. The Near-RT RIC **225** may be configured to include a logical function that enables near-real-time control and optimization of RAN elements and resources via data collection and actions over an interface (such as via an E**2** interface) connecting one or more CUs **210**, one or more DUs **230**, or both, as well as an O-eNB, with the Near-RT RIC **225**.
[0053] In some implementations, to generate AI/ML models to be deployed in the Near-RT RIC **225**, the Non-RT RIC **215** may receive parameters or external enrichment information from external servers. Such information may be utilized by the Near-RT RIC **225** and may be received at the SMO Framework **205** or the Non-RT RIC **215** from non-network data sources or from network functions. In some examples, the Non-RT RIC **215** or the Near-RT RIC **225** may be configured to tune RAN behavior or performance. For example, the Non-RT RIC **215** may monitor long-term trends and patterns for performance and employ AI/ML models to perform corrective actions through the SMO Framework **205** (such as reconfiguration via O**1**) or via creation of RAN management policies (such as A**1** policies).
[0054] FIG. **3** depicts aspects of an example BS **102** and a UE **104**.
[0055] Generally, BS **102** includes various processors (e.g., **320**, **330**, **338**, and **340**), antennas **334***a-t* (collectively **334**), transceivers **332***a-t* (collectively **332**), which include modulators and demodulators, and other aspects, which enable wireless transmission of data (e.g., data source **312**) and wireless reception of data (e.g., data sink **339**). For example, BS **102** may send and receive data between BS **102** and UE **104**. BS **102** includes controller/processor **340**, which may be configured to implement various functions described herein related to wireless communications.

[0056] Generally, UE **104** includes various processors (e.g., **358**, **364**, **366**, and **380**), antennas **352***a-r* (collectively **352**), transceivers **354***a-r* (collectively **354**), which include modulators and demodulators, and other aspects, which enable wireless transmission of data (e.g., retrieved from data source **362**) and wireless reception of data (e.g., provided to data sink **360**). UE **104** includes controller/processor **380**, which may be configured to implement various functions described herein related to wireless communications.
[0057] In regards to an example downlink transmission, BS **102** includes a transmit processor **320** that may receive data from a data source **312** and control information from a controller/processor **340**. The control information may be for the physical broadcast channel (PBCH), physical control format indicator channel (PCFICH), physical HARQ indicator channel (PHICH), physical downlink control channel (PDCCH), group common PDCCH (GC PDCCH), and/or others. The data may be for the physical downlink shared channel (PDSCH), in some examples.
[0058] Transmit processor **320** may process (e.g., encode and symbol map) the data and control information to obtain data symbols and control symbols, respectively. Transmit processor **320** may also generate reference symbols, such as for the primary synchronization signal (PSS), secondary synchronization signal (SSS), PBCH demodulation reference signal (DMRS), and channel state information reference signal (CSI-RS).
[0059] Transmit (TX) multiple-input multiple-output (MIMO) processor **330** may perform spatial processing (e.g., precoding) on the data symbols, the control symbols, and/or the reference symbols, if applicable, and may provide output symbol streams to the modulators (MODs) in transceivers **332***a*-**332***t*. Each modulator in transceivers **332***a*-**332***t* may process a respective output symbol stream to obtain an output sample stream. Each modulator may further process (e.g., convert to analog, amplify, filter, and upconvert) the output sample stream to obtain a downlink signal. Downlink signals from the modulators in transceivers **332***a*-**332***t* may be transmitted via the antennas **334***a*-**334***t*, respectively.
[0060] In order to receive the downlink transmission, UE **104** includes antennas **352***a*-**352***r* that may receive the downlink signals from the BS **102** and may provide received signals to the demodulators (DEMODs) in transceivers **354***a*-**354***r*, respectively. Each demodulator in transceivers **354***a*-**354***r* may condition (e.g., filter, amplify, downconvert, and digitize) a respective received signal to obtain input samples. Each demodulator may further process the input samples to obtain received symbols.
[0061] MIMO detector **356** may obtain received symbols from all the demodulators in transceivers **354***a*-**354***r*, perform MIMO detection on the received symbols if applicable, and provide detected symbols. Receive processor **358** may process (e.g., demodulate, deinterleave, and decode) the detected symbols, provide decoded data for the UE **104** to a data sink **360**, and provide decoded control information to a controller/processor **380**.
[0062] In regards to an example uplink transmission, UE **104** further includes a transmit processor **364** that may receive and process data (e.g., for the PUSCH) from a data source **362** and control information (e.g., for the physical uplink control channel (PUCCH)) from the controller/processor **380**. Transmit processor **364** may also generate reference symbols for a reference signal (e.g., for the sounding reference signal (SRS)). The symbols from the transmit processor **364** may be precoded by a TX MIMO processor **366** if applicable, further processed by the modulators in transceivers **354***a*-**354***r* (e.g., for SC-FDM), and transmitted to BS **102**.
[0063] At BS **102**, the uplink signals from UE **104** may be received by antennas **334***a-t*, processed by the demodulators in transceivers **332***a*-**332***t*, detected by a MIMO detector **336** if applicable, and further processed by a receive processor **338** to obtain decoded data and control information sent by UE **104**. Receive processor **338** may provide the decoded data to a data sink **339** and the decoded control information to the controller/processor **340**.
[0064] Memories **342** and **382** may store data and program codes for BS **102** and UE **104**, respectively.

[0065] Scheduler **344** may schedule UEs for data transmission on the downlink and/or uplink.
[0066] In various aspects, BS **102** may be described as transmitting and receiving various types of data associated with the methods described herein. In these contexts, "transmitting" may refer to various mechanisms of outputting data, such as outputting data from data source **312**, scheduler **344**, memory **342**, transmit processor **320**, controller/processor **340**, TX MIMO processor **330**, transceivers **332***a-t*, antenna **334***a-t*, and/or other aspects described herein. Similarly, "receiving" may refer to various mechanisms of obtaining data, such as obtaining data from antennas **334***a-t*, transceivers **332***a-t*, RX MIMO detector **336**, controller/processor **340**, receive processor **338**, scheduler **344**, memory **342**, and/or other aspects described herein.
[0067] In various aspects, UE **104** may likewise be described as transmitting and receiving various types of data associated with the methods described herein. In these contexts, "transmitting" may refer to various mechanisms of outputting data, such as outputting data from data source **362**, memory **382**, transmit processor **364**, controller/processor **380**, TX MIMO processor **366**, transceivers **354***a-t*, antenna **352***a-t*, and/or other aspects described herein. Similarly, "receiving" may refer to various mechanisms of obtaining data, such as obtaining data from antennas **352***a-t*, transceivers **354***a-t*, RX MIMO detector **356**, controller/processor **380**, receive processor **358**, memory **382**, and/or other aspects described herein.
[0068] In some aspects, one or more processors may be configured to perform various operations, such as those associated with the methods described herein, and transmit (output) to or receive (obtain) data from another interface that is configured to transmit or receive, respectively, the data.
[0069] FIGS. **4**A, **4**B, **4**C, and **4**D depict aspects of data structures for a wireless communications network, such as wireless communications network **100** of FIG. **1**.
[0070] In particular, FIG. **4**A is a diagram **400** illustrating an example of a first subframe within a 5G (e.g., 5G NR) frame structure, FIG. **4**B is a diagram **430** illustrating an example of DL channels within a 5G subframe, FIG. **4**C is a diagram **450** illustrating an example of a second subframe within a 5G frame structure, and FIG. **4**D is a diagram **480** illustrating an example of UL channels within a 5G subframe.
[0071] Wireless communications systems may utilize orthogonal frequency division multiplexing (OFDM) with a cyclic prefix (CP) on the uplink and downlink. Such systems may also support half-duplex operation using time division duplexing (TDD). OFDM and single-carrier frequency division multiplexing (SC-FDM) partition the system bandwidth (e.g., as depicted in FIGS. **4**B and **4**D) into multiple orthogonal subcarriers. Each subcarrier may be modulated with data. Modulation symbols may be sent in the frequency domain with OFDM and/or in the time domain with SC-FDM.
[0072] A wireless communications frame structure may be frequency division duplex (FDD), in which, for a particular set of subcarriers, subframes within the set of subcarriers are dedicated for either DL or UL. Wireless communications frame structures may also be time division duplex (TDD), in which, for a particular set of subcarriers, subframes within the set of subcarriers are dedicated for both DL and UL.
[0073] In FIG. **4**A and **4**C, the wireless communications frame structure is TDD where D is DL, U is UL, and X is flexible for use between DL/UL. UEs may be configured with a slot format through a received slot format indicator (SFI) (dynamically through DL control information (DCI), or semi-statically/statically through radio resource control (RRC) signaling). In the depicted examples, a 10 ms frame is divided into 10 equally sized 1 ms subframes. Each subframe may include one or more time slots. In some examples, each slot may include 7 or 14 symbols, depending on the slot format. Subframes may also include mini-slots, which generally have fewer symbols than an entire slot. Other wireless communications technologies may have a different frame structure and/or different channels.
[0074] In certain aspects, the number of slots within a subframe is based on a slot configuration and a numerology. For example, for slot configuration **0**, different numerologies (μ) **0** to **6** allow for 1,

2, 4, 8, 16, 32, and 64 slots, respectively, per subframe. For slot configuration **1**, different numerologies **0** to **2** allow for 2, 4, and 8 slots, respectively, per subframe. Accordingly, for slot configuration **0** and numerology μ, there are 14 symbols/slot and 2μ slots/subframe. The subcarrier spacing and symbol length/duration are a function of the numerology. The subcarrier spacing may be equal to 2.sup.μ×15 kHz, where μ is the numerology **0** to **6**. As such, the numerology μ=0 has a subcarrier spacing of 15 kHz and the numerology μ=6 has a subcarrier spacing of 960 kHz. The symbol length/duration is inversely related to the subcarrier spacing. FIGS. **4**A, **4**B, **4**C, and **4**D provide an example of slot configuration **0** with 14 symbols per slot and numerology μ=2 with 4 slots per subframe. The slot duration is 0.25 ms, the subcarrier spacing is 60 kHz, and the symbol duration is approximately 16.67 μs.
[0075] As depicted in FIGS. **4**A, **4**B, **4**C, and **4**D, a resource grid may be used to represent the frame structure. Each time slot includes a resource block (RB) (also referred to as physical RBs (PRBs)) that extends, for example, 12 consecutive subcarriers. The resource grid is divided into multiple resource elements (REs). The number of bits carried by each RE depends on the modulation scheme.
[0076] As illustrated in FIG. **4**A, some of the REs carry reference (pilot) signals (RS) for a UE (e.g., UE **104** of FIGS. **1** and **3**). The RS may include demodulation RS (DMRS) and/or channel state information reference signals (CSI-RS) for channel estimation at the UE. The RS may also include beam measurement RS (BRS), beam refinement RS (BRRS), and/or phase tracking RS (PT-RS).
[0077] FIG. **4**B illustrates an example of various DL channels within a subframe of a frame. The physical downlink control channel (PDCCH) carries DCI within one or more control channel elements (CCEs), each CCE including, for example, nine RE groups (REGs), each REG including, for example, four consecutive REs in an OFDM symbol.
[0078] A primary synchronization signal (PSS) may be within symbol **2** of particular subframes of a frame. The PSS is used by a UE (e.g., **104** of FIGS. **1** and **3**) to determine subframe/symbol timing and a physical layer identity.
[0079] A secondary synchronization signal (SSS) may be within symbol **4** of particular subframes of a frame. The SSS is used by a UE to determine a physical layer cell identity group number and radio frame timing.
[0080] Based on the physical layer identity and the physical layer cell identity group number, the UE can determine a physical cell identifier (PCI). Based on the PCI, the UE can determine the locations of the aforementioned DMRS. The physical broadcast channel (PBCH), which carries a master information block (MIB), may be logically grouped with the PSS and SSS to form a synchronization signal (SS)/PBCH block. The MIB provides a number of RBs in the system bandwidth and a system frame number (SFN). The physical downlink shared channel (PDSCH) carries user data, broadcast system information not transmitted through the PBCH such as system information blocks (SIBs), and/or paging messages.
[0081] As illustrated in FIG. **4**C, some of the REs carry DMRS (indicated as R for one particular configuration, but other DMRS configurations are possible) for channel estimation at the base station. The UE may transmit DMRS for the PUCCH and DMRS for the PUSCH. The PUSCH DMRS may be transmitted, for example, in the first one or two symbols of the PUSCH. The PUCCH DMRS may be transmitted in different configurations depending on whether short or long PUCCHs are transmitted and depending on the particular PUCCH format used. UE **104** may transmit sounding reference signals (SRS). The SRS may be transmitted, for example, in the last symbol of a subframe. The SRS may have a comb structure, and a UE may transmit SRS on one of the combs. The SRS may be used by a base station for channel quality estimation to enable frequency-dependent scheduling on the UL.
[0082] FIG. **4**D illustrates an example of various UL channels within a subframe of a frame. The PUCCH may be located as indicated in one configuration. The PUCCH carries uplink control

information (UCI), such as scheduling requests, a channel quality indicator (CQI), a precoding matrix indicator (PMI), a rank indicator (RI), and HARQ ACK/NACK feedback. The PUSCH carries data, and may additionally be used to carry a buffer status report (BSR), a power headroom report (PHR), and/or UCI.

Aspects Related to Improvements for Voice Call Continuity from WLAN to WWAN

[0083] Many wireless carriers are wireless local area network (WLAN)-preferred (e.g., WiFi-preferred) carriers as compared to wireless wide area network (WWAN)-preferred (e.g., cellular-preferred). In other words, these carriers prefer that voice calls be made over WLAN, as compared to over cellular or WWAN services. Under WLAN coverage, user equipments (UEs) would be required to camp on a WLAN access point (AP) for making and receiving calls. With fifth generation new radio (5G NR) standalone (SA) becoming increasingly commercialized by major cellular operators, internet protocol multimedia subsystem (IMS) calls are becoming possible over 5G NR, long term evolution (LTE), and WLAN (e.g., WiFi).

[0084] In some cases, certain original equipment manufacturers (OEMs) have configured their devices (e.g., UEs) with a preference for LTE evolved packet system fallback (EPSFB) for voice calls rather than voice over new radio (VoNR) due to equality concerns and device capability. In other words, when making voice calls, these devices (e.g., UEs) may be configured to fall back from a 5G NR radio access technology (RAT) network to an LTE RAT network due to quality concerns and device compatibility issues. In some cases, however, the EPSFB may cause certain issues with voice calls over a WLAN RAT network, such as call muting and call dropping.

[0085] FIG. **5** is a call flow diagram including operations **500** that illustrate the call muting issue that may arise when an EPSFB-based UE performs a voice call over a WLAN RAN, such as WiFi. For example, as shown, operations **500** illustrated in FIG. **5** involve a UE **502**, a 5G network entity **504**, an LTE network entity **506**, and a WLAN AP **508**. In some cases, the UE **502** may be capable of performing WLAN-based voice calls and EPSFB-based voice calls. In other words, the UE **502** may perform voice calls over a WLAN RAT network (e.g., a WiFi network) or, when using cellular- or WWAN-based service, may be required to fall back to an LTE RAT network to perform a voice call.

[0086] As illustrated, operations **500** begin at **510** with the UE **502** performing IMS registration with the 5G network entity **504** for data services using a 5G RAT network.

[0087] Thereafter, at **512**, the UE **502** may establish a connection with the WLAN AP **508** using the WLAN RAT network.

[0088] Thereafter, as shown at **514**, the UE **502** initiates and performs a voice call via the WLAN AP **508**.

[0089] At **516**, during the voice call, the UE **502** detects that a quality or strength of the connection with the WLAN AP **508** (e.g., a signal strength or quality of the WLAN RAT network) has dropped below a particular threshold, significantly impacting a quality of the voice call.

[0090] In response to detecting that the quality or strength of the connection with the WLAN AP **508** has dropped below the particular threshold, at **518**, the UE **502** transmits an IMS PDN handover request to the 5G network entity **504**, requesting that the UE **502** be handed over to the 5G network entity **504** to continue the voice call. It should be appreciated that, even though the UE **502** is an EPSFB-based UE, the UE **502** transmits the IMS PDN handover request to the 5G network entity **504** due to the 5G RAT network having priority over the LTE RAT network.

[0091] Thereafter, as shown at **520**, because the UE **502** is an EPSFB-based UE, the 5G network entity **504** redirects the UE **502** to the LTE network entity **506** based on the IMS PDN handover request.

[0092] As shown at **522**, the UE **502** transmits a tracking area update (TAU) request message to the LTE network entity **506**. The UE **502** then receives a TAU accept message from the LTE network entity **506** at **524**.

[0093] Thereafter, as shown at **526**, the UE **502** transmits another IMS PDN handover request to the LTE network entity **506**, requesting a handover from the WLAN AP **508** to LTE network entity **506** in order to continue the voice call that was initiated with the WLAN AP **508**.

[0094] At **528**, the UE **502** then receives a PDN handover accept message from the LTE network entity **506** based on the IMS PDN handover request.

[0095] At **530**, the UE **502** may continue the voice call over the LTE RAT network after the handover from WIFI to LTE is complete.

[0096] As can be seen in FIG. **5**, when the quality or strength of the WLAN connection deteriorates below a threshold during a voice call being performed by an EPSFB-based UE, the process for reestablishing or continuing that voice call involves multiple handovers, which causes the voice call to be muted (e.g., no voice data is able to be communicated) for a significant amount of time. This muting issue, in turn, leads to poor user experience at the UE.

[0097] As noted above, EPSFB may also cause call dropping issues with VoWIFI calls. FIG. **6** is a call flow diagram including operations **600** that illustrate the call muting issue that may arise when an EPSFB-based UE performs a voice call over a WLAN RAT network, such as WiFi. For example, as shown, operations **600** illustrated in FIG. **6** involve a UE **602**, a 5G network entity **604**, an LTE network entity **606**, and a WLAN AP **608**. In some cases, the UE **602** may be capable of performing WLAN-based voice calls and EPSFB-based voice calls. In other words, the UE **602** may perform voice calls over a WLAN RAT network (e.g., WiFi) or, when using cellular- or WWAN-based service, may be required to fall back to an LTE RAT network to perform a voice call.

[0098] As illustrated, operations **600** begin at **610** with the UE **602** performing IMS registration with the 5G network entity **604** for data services using a 5G RAT network.

[0099] Thereafter, at **612**, the UE **602** may establish a connection with the WLAN AP **608** using the WLAN RAT network.

[0100] Thereafter, as shown at **614**, the UE **602** initiates and performs a voice call via the WLAN AP **608**.

[0101] At **616**, during the voice call, the UE **602** detects that a quality or strength of the connection with the WLAN AP **608** (e.g., a signal strength or quality of the WLAN RAT network) has dropped below a particular threshold, significantly impacting a quality of the voice call.

[0102] In response to detecting that the quality or strength of the connection with the WLAN AP **608** has dropped below the particular threshold, at **618**, the UE **602** transmits an IMS PDN handover request to the 5G network entity **604**, requesting that the UE **602** be handed over to the 5G network entity **604** to continue the voice call. As noted above, it should be appreciated that, even though the UE **602** is an EPSFB-based UE, the UE **602** transmits the IMS PDN handover request to the 5G network entity **604** due to the 5G RAT network having priority over the LTE RAT network.

[0103] In some cases, however, as shown at **620**, due to certain issues at the 5G network entity **604**, instead of redirecting the UE **602** over to the LTE network entity **606**, the 5G network entity **604** instead transmits an IMS PDN handover reject message to the UE **502**, indicating that the handover to the 5G network entity **604** has failed. In some cases, receiving the IMS PDN handover reject message from the 5G network entity **604** may lead to the UE **602** remaining “stuck” on the WLAN RAT network since the UE **602** will continue to prioritize the 5G RAT network over the LTE RAT network, leading to the UE **602** continually trying to handover to the 5G network entity **604** and being rejected. This process may continue indefinitely until the connection established with the WLAN AP **608** fails and the voice call is dropped. For example, as shown at **622**, as a result of the handover failing and not being redirected to the LTE network entity **606**, the voice call may drop or fail, leading to poor user experience.

[0104] Accordingly aspects of the present disclosure provide techniques to help alleviate the issues of call muting and call dropping that may arise when an EPSFB-based UE performs a voice call over a WLAN RAT network. For example, in some cases, to help alleviate the call muting issue described above with respect to FIG. **5**, these techniques may involve the UE performing a reselection procedure during the voice call to reselect the LTE RAT network prior to a WLAN connection deteriorating below a threshold. In some cases, the UE may perform the reselection procedure to reselect to the LTE RAT network irrespective of a relative priority assigned to the LTE RAT network. By performing the reselection procedure to reselect to the LTE RAT network prior to the WLAN connection deteriorating below the threshold, the UE may avoid having to perform the handover from a WLAN AP to a 5G network entity. In other words, by performing the reselection procedure to reselect to the LTE RAT network prior to the WLAN connection deteriorating below the threshold, the UE can avoid performing at least operations **518**, **520**, **522**, and **524** to reestablish the voice call when the WLAN connection deteriorates below the threshold. This may reduce the amount of time it takes to reestablish the voice call when the WLAN connection deteriorates below the threshold, thereby avoiding the negative impacts to user experience described above.

[0105] In other cases, to help alleviate the call dropping issues describe above with respect to FIG. **6**, the techniques provided herein may involve the UE de-prioritizing a 5G RAT network and attempting a handover to an LTE network entity when the UE receives a handover reject message from a 5G network entity on the 5G RAT network. This may avoid the scenario in which the UE is effectively stuck on the WLAN RAT network and unable to handover to the LTE network entity due to the 5G RAT network having priority over the LTE RAT network, as described above. For example, as noted, by de-prioritizing a 5G RAT network, the UE may then be free to attempt a handover to an LTE network entity to reestablish and continue the voice call.

Example Operations of Entities in a Communications Network

[0106] FIG. **7** depicts a process flow including operations **700** for communications in a network between a UE **702**, a 5G network entity **704**, an LTE network entity **706**, and a WLAN AP **708**. In some aspects, the 5G network entity **704** and the LTE network entity **706** may be examples of the BS **102** depicted and described with respect to FIGS. **1** and **3** or a disaggregated base station depicted and described with respect to FIG. **2**. Similarly, the UE **702** may be an example of UE **104** depicted and described with respect to FIGS. **1** and **3**. In some cases, the WLAN AP **708** may be an example of a WiFi access point. Additionally, in some cases, the UE **702** may be capable of performing WLAN-based voice calls or sessions and EPSFB-based voice calls or sessions. In other words, the UE **702** may perform voice calls over a WLAN RAT network (e.g., WiFi) or, when using cellular- or WWAN-based service, may be required to fall back to an LTE RAT network to perform a voice call.

[0107] As illustrated, operations **700** begin at **710** with the UE **702** performing IMS registration with the 5G network entity **704** for data services using a 5G RAT network.

[0108] Thereafter, at **712**, the UE **702** may establish a connection with the WLAN AP **708** using the WLAN RAT network.

[0109] Thereafter, as shown at **714**, the UE **702** initiates and performs a voice call via the WLAN AP **708**.

[0110] Thereafter, as shown at **716**, prior to a quality or strength of the connection with the WLAN AP **708** (e.g., a signal strength or quality of the WLAN RAT network) has dropping below a particular threshold, the UE **702** may perform reselection to the LTE network entity **706**. For example, the UE **702** may reselect to the LTE network entity **706** autonomously (e.g., without involvement of the 5G RAT network) and may, thereafter, initiate a TAU or registration procedure to inform the LTE RAT network that the UE **702** is camped on the LTE network entity **706**.

[0111] Thereafter, at **718**, during the voice call, the UE **702** may detect that the quality or the strength of the connection with the WLAN AP **708** (e.g., a signal strength or quality of the WLAN RAT network) has dropped below the particular threshold, significantly impacting a quality of the voice call. For example, in some cases, the UE **702** may detect that a received signal strength indicator (RSSI) associated with the connection with the WLAN AP **708** has fallen below an RSSI threshold.

[0112] In such cases, at **720**, rather than first sending a handover request to the 5G network entity **704**, the UE **702** may instead send an IMS PDN handover request (e.g., an EPSFB-based handover) to the LTE network entity **706** based the reselection that the UE **702** performed at **716**. Accordingly, at **722**, based on the IMS PDN handover request, the UE **702** may receive an IMS PDN handover accept message from the LTE network entity **706**, allowing the UE **702** to be handed over to the LTE network entity **706**.

[0113] Thereafter, as shown at **724**, the voice call may be reestablished and may continue via the LTE network entity **706** using the LTE RAT network. As noted above, by performing reselection to the LTE network entity **706** prior to the quality or strength of the connection with the WLAN AP **708** deteriorating below the threshold, the UE **702** may be able to more quickly reestablish the voice call using the LTE RAT network, thereby reducing negative effects to user experience at the UE **702**.

[0114] Thereafter, as shown at **726**, the UE **702** may end the voice call.

[0115] Thereafter, as shown at **728**, once the voice call has ended, the UE **702** may perform reselection back to the 5G network entity **704**.

[0116] FIG. **8** depicts a process flow including operations **800** for communications in a network between a UE **802**, a 5G network entity **804**, an LTE network entity **806**, and a WLAN AP **808**. In some aspects, the 5G network entity **804** and the LTE network entity **806** may be examples of the BS **102** depicted and described with respect to FIGS. **1** and **3** or a disaggregated base station depicted and described with respect to FIG. **2**. Similarly, the UE **802** may be an example of UE **104** depicted and described with respect to FIGS. **1** and **3**. In some cases, the WLAN AP **808** may be an example of a WiFi access point. Additionally, in some cases, the UE **802** may be capable of performing WLAN-based voice calls or sessions and EPS FB-based voice calls or sessions. In other words, the UE **602** may perform voice calls over a WLAN RAT network (e.g., WiFi) or, when using cellular- or WWAN-based service, may be required to fall back to an LTE RAT network to perform a voice call.

[0117] As illustrated, operations **800** begin at **810** with the UE **802** performing IMS registration with the 5G network entity **804** for data services using a 5G RAT network.

[0118] Thereafter, at **812**, the UE **802** may establish a connection with the WLAN AP **808** using the WLAN RAT network.

[0119] Thereafter, as shown at **814**, the UE **802** initiates and performs a voice call via the WLAN AP **808**.

[0120] At **816**, during the voice call, the UE **802** detects that a quality or strength of the connection with the WLAN AP **808** (e.g., a signal strength or quality of the WLAN RAT network) has dropped below a particular threshold, significantly impacting a quality of the voice call. For example, in some cases, the UE **802** may detect that an RSSI associated with the connection with the WLAN AP **808** has fallen below an RSSI threshold.

[0121] In response to detecting that the quality or strength of the connection with the WLAN AP **808** has dropped below the particular threshold, at **818**, the UE **802** transmits an IMS PDN handover request to the 5G network entity **804**, requesting that the UE **802** be handed over to the 5G network entity **804** to continue the voice call. As noted above, it should be appreciated that, even though the UE **802** is an EPSFB-based UE, the UE **802** transmits the IMS PDN handover request to the 5G network entity **804** due to the 5G RAT network having priority over the LTE RAT network.

[0122] In some cases, however, as shown at **820**, due to certain issues at the 5G network entity **804**, instead of redirecting the UE **802** over to the LTE network entity **806**, the 5G network entity **804** instead transmits an IMS PDN handover reject message to the UE **502**, indicating that the handover to the 5G network entity **804** has failed.

[0123] In order to avoid a scenario in which the UE **802** is “stuck” on the WLAN RAT network due to the 5G RAT network having a higher priority than the LTE RAT network and the handover to the

5G network entity **804** failing (e.g., as described above with respect to FIG. **6**), the UE **802** may de-prioritize the 5G RAT network at **822** and instead transmit a IMS PDN handover request to the LTE network entity **806** at **824**. As shown at **826**, the UE **802** receives an IMS PDN handover accept message from the LTE network entity **806**, allowing the UE **802** to be handed over to the LTE network entity **806**.

[0124] Thereafter, as shown at **828**, the voice call may be reestablished and may continue via the LTE network entity **806** using the LTE RAT network. As noted above, by de-prioritizing the 5G RAT network and instead performing the handover to the LTE network entity **806** (e.g., even though the 5G RAT network has a higher relative priority than the LTE RAT network), the UE **802** is able to avoid scenarios in which the voice call drops due to the UE **802** being "stuck" on the WLAN RAT network, thereby reducing negative effects to user experience at the UE **802**.

[0125] Thereafter, as shown at **830**, the UE **802** may end the voice call.

[0126] Thereafter, as shown at **832**, once the voice call has ended, the UE **802** may perform reselection back to the 5G network entity **804**.

[0127] It should be appreciated that the operations illustrated in FIGS. **7** and **8** may be applicable to other types of network entities associated with other types of RAT networks, such as a network entities associated with a sixth generation (6G) RAT network, network entities associated with a third generation (3G) RAT network, and the like. As an example, in some cases, the LTE network entity **706** may instead by a network entity associated with a 5G RAT network and the 5G network entity **704** may be a network entity associated with a 6G RAT network.

Example Operations of a User Equipment

[0128] FIG. **9** shows an example of a method **900** of wireless communications at a user equipment (UE), such as a UE **104** of FIGS. **1** and **3**.

[0129] Method **900** begins at step **905** with performing, while a session is active on a first radio access technology (RAT) network, reselection to a second RAT network from a third RAT network. In some cases, the operations of this step refer to, or may be performed by, circuitry for performing and/or code for performing as described with reference to FIG. **10**.

[0130] In some aspects, the session may comprise a voice call and the UE is capable of supporting evolved packet system (EPS) fallback for the voice call. In some aspects, the first RAT network comprises a WLAN, such as a WiFi network or another type of WLAN. In some aspects, the second and third RAT networks comprise wireless wide area networks (WWANs). For example, in some aspects, the second RAT network comprises an LTE network and third RAT network comprises a 5G NR network.

[0131] Method **900** then proceeds to step **910** with performing a handover to the second RAT network, after detecting a condition impacting the session on the first RAT network. In some cases, the operations of this step refer to, or may be performed by, circuitry for performing and/or code for performing as described with reference to FIG. **10**.

[0132] Method **900** then proceeds to step **915** with resuming the session on the second RAT network. For example, in some aspects resuming the session on the second RAT network may include resuming the voice call on the second RAT network. In some cases, the operations of this step refer to, or may be performed by, circuitry for resuming and/or code for resuming as described with reference to FIG. **10**.

[0133] In some aspects, the condition involves signal quality in the first RAT network. For example, in some aspects, the condition may include the signal quality in the first RAT network falling below a threshold.

[0134] In some aspects, performing the handover comprises transmitting a packet data network (PDN) handover request to the second RAT network. For example, in some aspects, transmitting the handover request to the second RAT network may include transmitting the handover request to a network entity of the second RAT network.

[0135] In some aspects, the session is resumed on the second RAT network after receiving an acceptance of the PDN handover request.

[0136] In some aspects, the method **900** further includes performing reselection to a third RAT network after an end of the session on the second RAT network. In some cases, the operations of this step refer to, or may be performed by, circuitry for performing and/or code for performing as described with reference to FIG. **10**.

[0137] In some aspects, the UE is camped on the third RAT network when it performs the reselection to the second RAT network.

[0138] In some aspects, the UE performs the reselection to the second RAT network irrespective of a relative priority assigned to the third RAT and second RAT.

[0139] In one aspect, method **900**, or any aspect related to it, may be performed by an apparatus, such as communications device **1000** of FIG. **10**, which includes various components operable, configured, or adapted to perform the method **900**. Communications device **1000** is described below in further detail.

[0140] Note that FIG. **9** is just one example of a method, and other methods including fewer, additional, or alternative steps are possible consistent with this disclosure.

Example Communications Devices

[0141] FIG. **10** depicts aspects of an example communications device **1000**. In some aspects, communications device **1000** is a user equipment, such as UE **104** described above with respect to FIGS. **1** and **3**.

[0142] The communications device **1000** includes a processing system **1005** coupled to the transceiver **1055** (e.g., a transmitter and/or a receiver). The transceiver **1055** is configured to transmit and receive signals for the communications device **1000** via the antenna **1060**, such as the various signals as described herein. The processing system **1005** may be configured to perform processing functions for the communications device **1000**, including processing signals received and/or to be transmitted by the communications device **1000**.

[0143] The processing system **1005** includes one or more processors **1010**. In various aspects, the one or more processors **1010** may be representative of one or more of receive processor **358**, transmit processor **364**, TX MIMO processor **366**, and/or controller/processor **380**, as described with respect to FIG. **3**. The one or more processors **1010** are coupled to a computer-readable medium/memory **1030** via a bus **1050**. In certain aspects, the computer-readable medium/memory **1030** is configured to store instructions (e.g., computer-executable code) that when executed by the one or more processors **1010**, cause the one or more processors **1010** to perform the method **900** described with respect to FIG. **9**, or any aspect related to it. Note that reference to a processor performing a function of communications device **1000** may include one or more processors **1010** performing that function of communications device **1000**.

[0144] In the depicted example, computer-readable medium/memory **1030** stores code (e.g., executable instructions), such as code for performing **1035**, code for resuming **1040**, and code for transmitting **1045**. Processing of the code for performing **1035**, code for resuming **1040**, and code for transmitting **1045** may cause the communications device **1000** to perform the method **900** described with respect to FIG. **9**, or any aspect related to it.

[0145] The one or more processors **1010** include circuitry configured to implement (e.g., execute) the code stored in the computer-readable medium/memory **1030**, including circuitry such as circuitry for performing **1015**, circuitry for resuming **1020**, and circuitry for transmitting **1025**. Processing with circuitry for performing **1015**, circuitry for resuming **1020**, and circuitry for transmitting **1025** may cause the communications device **1000** to perform the method **900** described with respect to FIG. **9**, or any aspect related to it.

[0146] Various components of the communications device **1000** may provide means for performing the method **900** described with respect to FIG. **9**, or any aspect related to it. For example, means for transmitting, sending or outputting for transmission may include transceivers **354** and/or antenna(s) **352** of the UE **104** illustrated in FIG. **3** and/or the transceiver **1055** and the antenna **1060** of the communications device **1000** in FIG. **10**. Means for receiving or obtaining may include transceivers **354** and/or antenna(s) **352** of the UE **104** illustrated in FIG. **3** and/or the transceiver **1055** and the antenna **1060** of the communications device **1000** in FIG. **10**.

Example Clauses

[0147] Implementation examples are described in the following numbered clauses:

[0148] Clause 1: A method for wireless communications at a user equipment (UE), comprising: performing, while a session is active on a first radio access technology (RAT) network, reselection to a second RAT network from a third RAT network; performing a handover to the second RAT network, after detecting a condition impacting the session on the first RAT network; and resuming the session on the second RAT network.

[0149] Clause 2: The method of Clause 1, wherein the condition involves signal quality in the first RAT network.

[0150] Clause 3: The method of any one of Clauses 1-2, wherein performing the handover comprises transmitting a packet data network (PDN) handover request to the second RAT network.

[0151] Clause 4: The method of Clause 3, wherein the session is resumed on the second RAT network after receiving an acceptance of the PDN handover request.

[0152] Clause 5: The method of any one of Clauses 1-4, further comprising performing reselection to a third RAT network after an end of the session on the second RAT network.

[0153] Clause 6: The method of Clause 5, wherein the UE is camped on the third RAT network when it performs the reselection to the second RAT network.

[0154] Clause 7: The method of Clause 5, wherein the UE performs the reselection to the second RAT network irrespective of a relative priority assigned to the third RAT and second RAT.

[0155] Clause 8: The method of Clause 5, wherein: the session comprises a voice call; and the UE is capable of supporting evolved packet system (EPS) fallback for the voice call.

[0156] Clause 9: The method of Clause 5, wherein: the first RAT network comprises a wireless local area network (WLAN); and the second and third RAT networks comprise wireless wide area networks (WWANs).

[0157] Clause 10: The method of Clause 9, wherein: the second RAT network comprises a long term evolution (LTE) network; and the third RAT network comprises a new radio (NR) network.

[0158] Clause 11: An apparatus, comprising: at least one memory comprising executable instructions; and at least one processor configured to execute the executable instructions and cause the apparatus to perform a method in accordance with any one of Clauses 1-10.

[0159] Clause 12: An apparatus, comprising means for performing a method in accordance with any one of Clauses 1-10.

[0160] Clause 13: A non-transitory computer-readable medium comprising executable instructions that, when executed by at least one processor of an apparatus, cause the apparatus to perform a method in accordance with any one of Clauses 1-10.

[0161] Clause 14: A computer program product embodied on a computer-readable storage medium comprising code for performing a method in accordance with any one of Clauses 1-10.

Additional Considerations

[0162] The preceding description is provided to enable any person skilled in the art to practice the various aspects described herein. The examples discussed herein are not limiting of the scope, applicability, or aspects set forth in the claims. Various modifications to these aspects will be readily apparent to those skilled in the art, and the general principles defined herein may be applied to other aspects. For example, changes may be made in the function and arrangement of elements discussed without departing from the scope of the disclosure. Various examples may omit, substitute, or add various procedures or components as appropriate. For instance, the methods described may be performed in an order different from that described, and various actions may be added, omitted, or combined. Also, features described with respect to some examples may be combined in some other examples. For example, an apparatus may be implemented or a method may be practiced using any number of the aspects set forth herein. In addition, the scope of the disclosure is intended to cover such an apparatus or method that is practiced using other structure, functionality, or structure and functionality in addition to, or other than, the various aspects of the disclosure set forth herein. It should be understood that any aspect of the disclosure disclosed herein may be embodied by one or more elements of a claim.

[0163] The various illustrative logical blocks, modules and circuits described in connection with the present disclosure may be implemented or performed with a general purpose processor, a digital signal processor (DSP), an ASIC, a field programmable gate array (FPGA) or other programmable logic device (PLD), discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any commercially available processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, a system on a chip (SoC), or any other such configuration.

[0164] As used herein, “a processor,” “at least one processor” or “one or more processors” generally refers to a single processor configured to perform one or multiple operations or multiple processors configured to collectively perform one or more operations. In the case of multiple processors, performance of the one or more operations could be divided amongst different processors, though one processor may perform multiple operations, and multiple processors could collectively perform a single operation. Similarly, “a memory,” “at least one memory” or “one or more memories” generally refers to a single memory configured to store data and/or instructions, multiple memories configured to collectively store data and/or instructions.

[0165] As used herein, a phrase referring to “at least one of” a list of items refers to any combination of those items, including single members. As an example, “at least one of: a, b, or c” is intended to cover a, b, c, a-b, a-c, b-c, and a-b-c, as well as any combination with multiples of the same element (e.g., a-a, a-a-a, a-a-b, a-a-c, a-b-b, a-c-c, b-b, b-b-b, b-b-c, c-c, and c-c-c or any other ordering of a, b, and c).

[0166] As used herein, the term “determining” encompasses a wide variety of actions. For example, “determining” may include calculating, computing, processing, deriving, investigating, looking up (e.g., looking up in a table, a database or another data structure), ascertaining and the like. Also, “determining” may include receiving (e.g., receiving information), accessing (e.g., accessing data in a memory) and the like. Also, “determining” may include resolving, selecting, choosing, establishing and the like.

[0167] The methods disclosed herein comprise one or more actions for achieving the methods. The method actions may be interchanged with one another without departing from the scope of the claims. In other words, unless a specific order of actions is specified, the order and/or use of specific actions may be modified without departing from the scope of the claims. Further, the various operations of methods described above may be performed by any suitable means capable of performing the corresponding functions. The means may include various hardware and/or software component(s) and/or module(s), including, but not limited to a circuit, an application specific integrated circuit (ASIC), or processor.

[0168] The following claims are not intended to be limited to the aspects shown herein, but are to be accorded the full scope consistent with the language of the claims. Within a claim, reference to an element in the singular is not intended to mean “one and only one” unless specifically so stated, but rather “one or more.” Unless specifically stated otherwise, the term “some” refers to one or more. No claim element is to be construed under the provisions of 35 U.S.C. § 112(f) unless the element is expressly recited using the phrase “means for”. All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims.

Claims

1. An apparatus for wireless communication at a user equipment (UE), comprising: one or more processors configured to execute instructions stored on one or more memories and to cause the UE to: perform, while a session is active on a first radio access technology (RAT) network, reselection to a second RAT network from a third RAT network; perform a handover to the second RAT network, after detecting a condition impacting the session on the first RAT network; and resume the session on the second RAT network.
2. The apparatus of claim 1, wherein the condition involves signal quality in the first RAT network.
3. The apparatus of claim 1, wherein in order to perform the handover, the one or more processors are further configured to cause the UE to transmit a packet data network (PDN) handover request to the second RAT network.
4. The apparatus of claim 3, wherein the session is resumed on the second RAT network after receiving an acceptance of the PDN handover request.
5. The apparatus of claim 1, wherein the one or more processors are further configured to cause the UE to perform reselection to a third RAT network after an end of the session on the second RAT network.
6. The apparatus of claim 5, wherein the one or more processors are configured to cause the UE to performs the reselection to the second RAT network when the UE is camped on the third RAT network.
7. The apparatus of claim 5, wherein the one or more processors are configured to cause the UE to perform the reselection to the second RAT network irrespective of a relative priority assigned to the third RAT and second RAT.
8. The apparatus of claim 5, wherein: the session comprises a voice call; and the UE is capable of supporting evolved packet system (EPS) fallback for the voice call.
9. The apparatus of claim 5, wherein: the first RAT network comprises a wireless local area network (WLAN); and the second and third RAT networks comprise wireless wide area networks (WWANs).
10. The apparatus of claim 9, wherein: the second RAT network comprises a long term evolution (LTE) network; and the third RAT network comprises a new radio (NR) network.
11. A method for wireless communications at a user equipment (UE), comprising: performing, while a session is active on a first radio access technology (RAT) network, reselection to a second RAT network from a third RAT network; performing a handover to the second RAT network, after detecting a condition impacting the session on the first RAT network; and resuming the session on the second RAT network.
12. The method of claim 11, wherein the condition involves signal quality in the first RAT network.
13. The method of claim 11, wherein performing the handover comprises transmitting a packet data network (PDN) handover request to the second RAT network.
14. The method of claim 13, wherein the session is resumed on the second RAT network after receiving an acceptance of the PDN handover request.
15. The method of claim 11, further comprising performing reselection to a third RAT network after an end of the session on the second RAT network.
16. The method of claim 15, wherein the UE is camped on the third RAT network when it performs the reselection to the second RAT network.
17. The method of claim 15, wherein the UE performs the reselection to the second RAT network irrespective of a relative priority assigned to the third RAT and second RAT.

18. The method of claim 15, wherein: the session comprises a voice call; and the UE is capable of supporting evolved packet system (EPS) fallback for the voice call.
19. The method of claim 15, wherein: the first RAT network comprises a wireless local area network (WLAN); the second and third RAT networks comprise wireless wide area networks (WWANs); the second RAT network comprises a long term evolution (LTE) network; and the third RAT network comprises a new radio (NR) network.
20. A non-transitory computer-readable medium for wireless communications, comprising: instructions that, when executed by one or more processors of a user equipment (UE), cause the UE to: perform, while a session is active on a first radio access technology (RAT) network, reselection to a second RAT network from a third RAT network; perform a handover to the second RAT network, after detecting a condition impacting the session on the first RAT network; and resume the session on the second RAT network.